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(54) **ATHLETIC SOCK**

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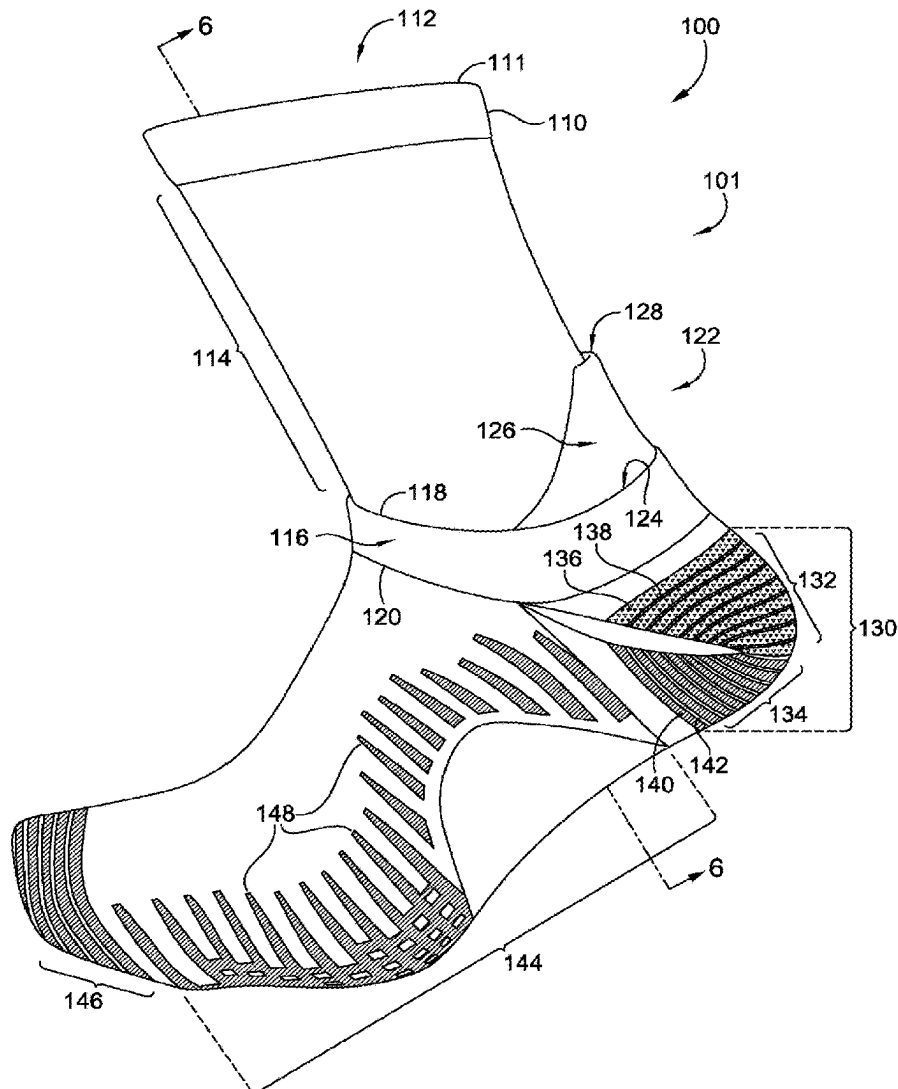
Related U.S. Application Data

(63) Continuation of application No. 18/530,718, filed on Dec. 6, 2023, now Pat. No. 12,262,758, which is a continuation of application No. 17/525,640, filed on Nov. 12, 2021, now Pat. No. 11,882,886.

(60) Provisional application No. 63/157,890, filed on Mar. 8, 2021, provisional application No. 63/164,677, filed

(57) **ABSTRACT**

Aspects herein are directed to a knit sock having a leg portion and an integrally knit ankle welt extending around a circumference of the leg portion and positioned at a lower end of the leg portion. The knit sock further includes an integrally knit tab located at a back aspect of the ankle welt.



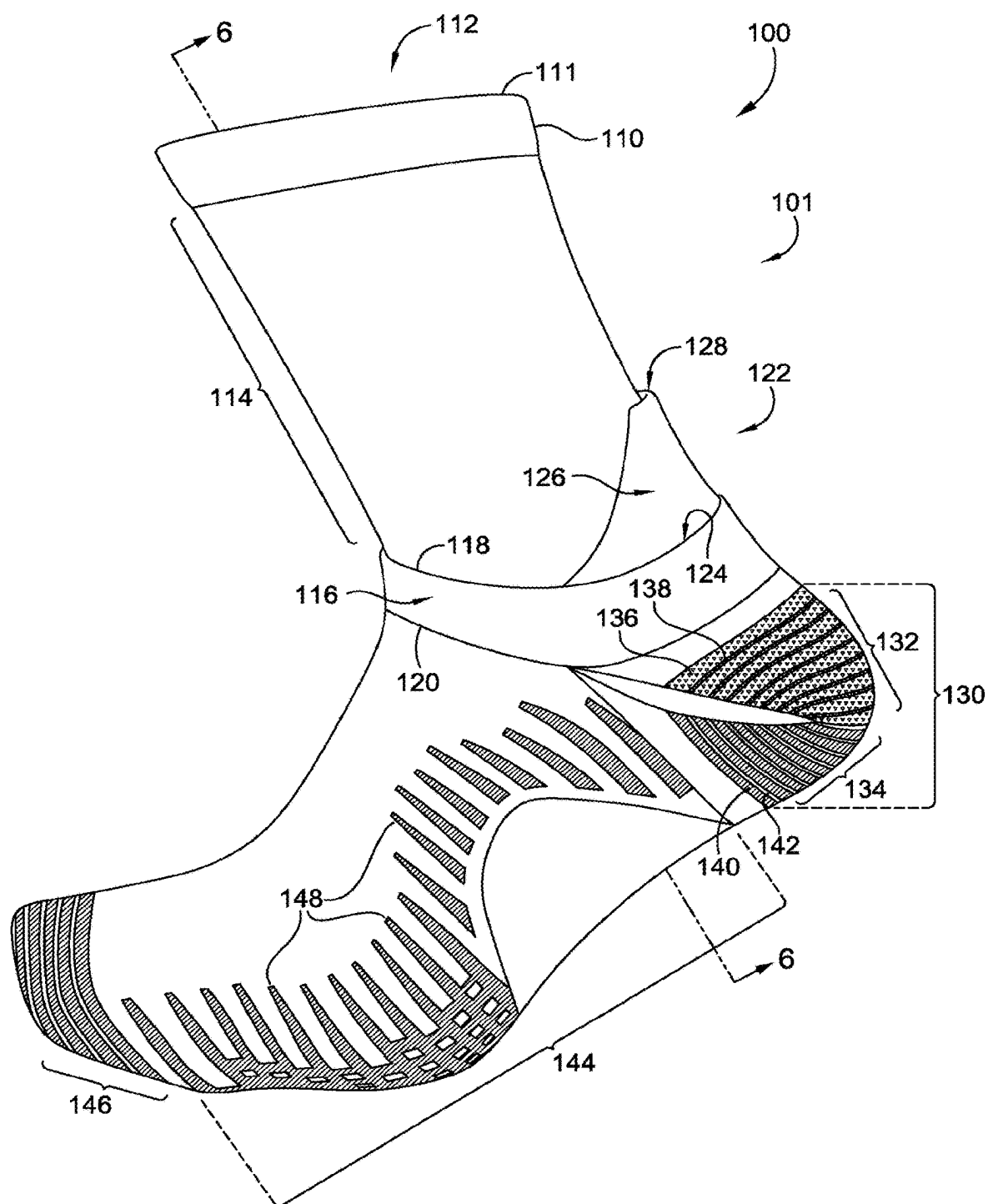
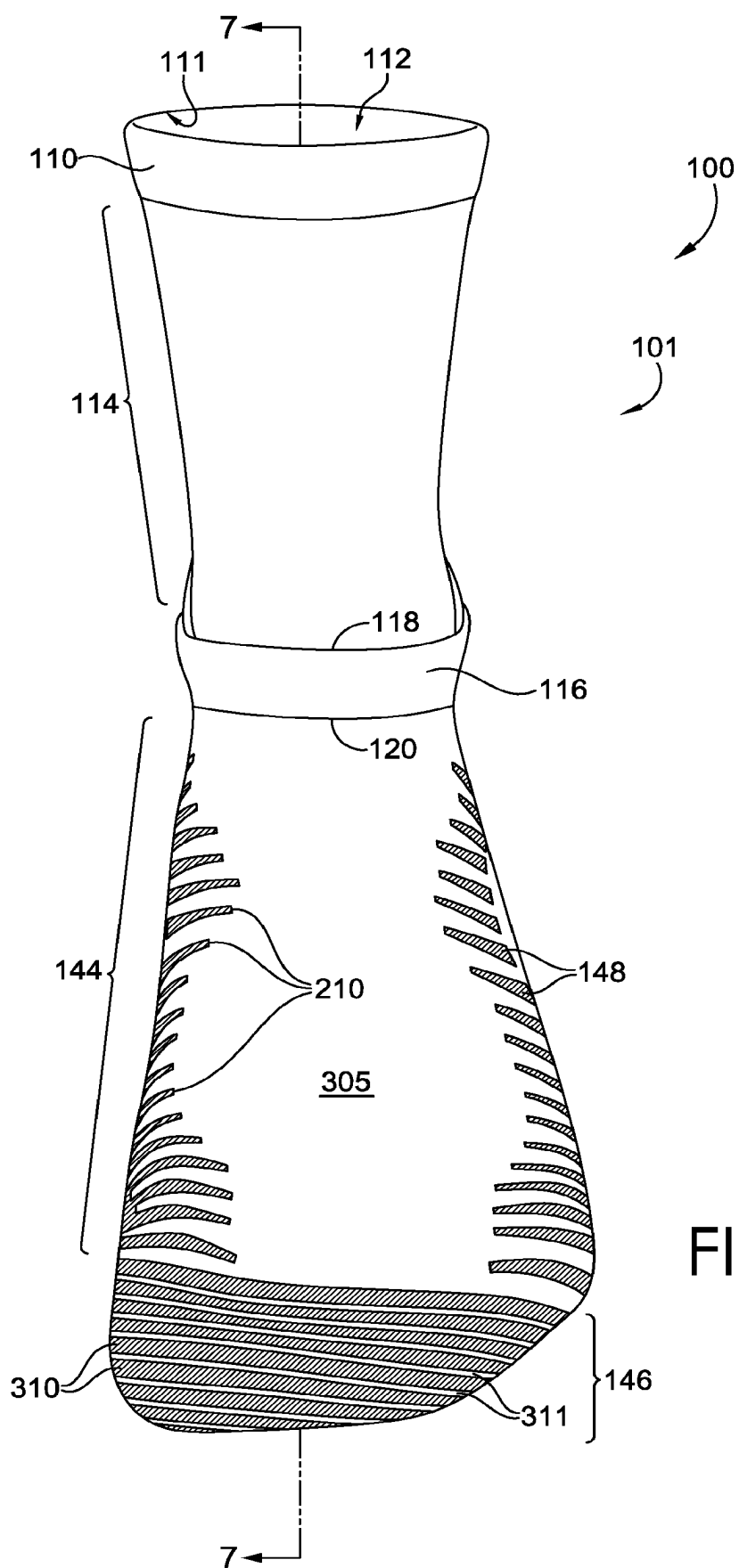
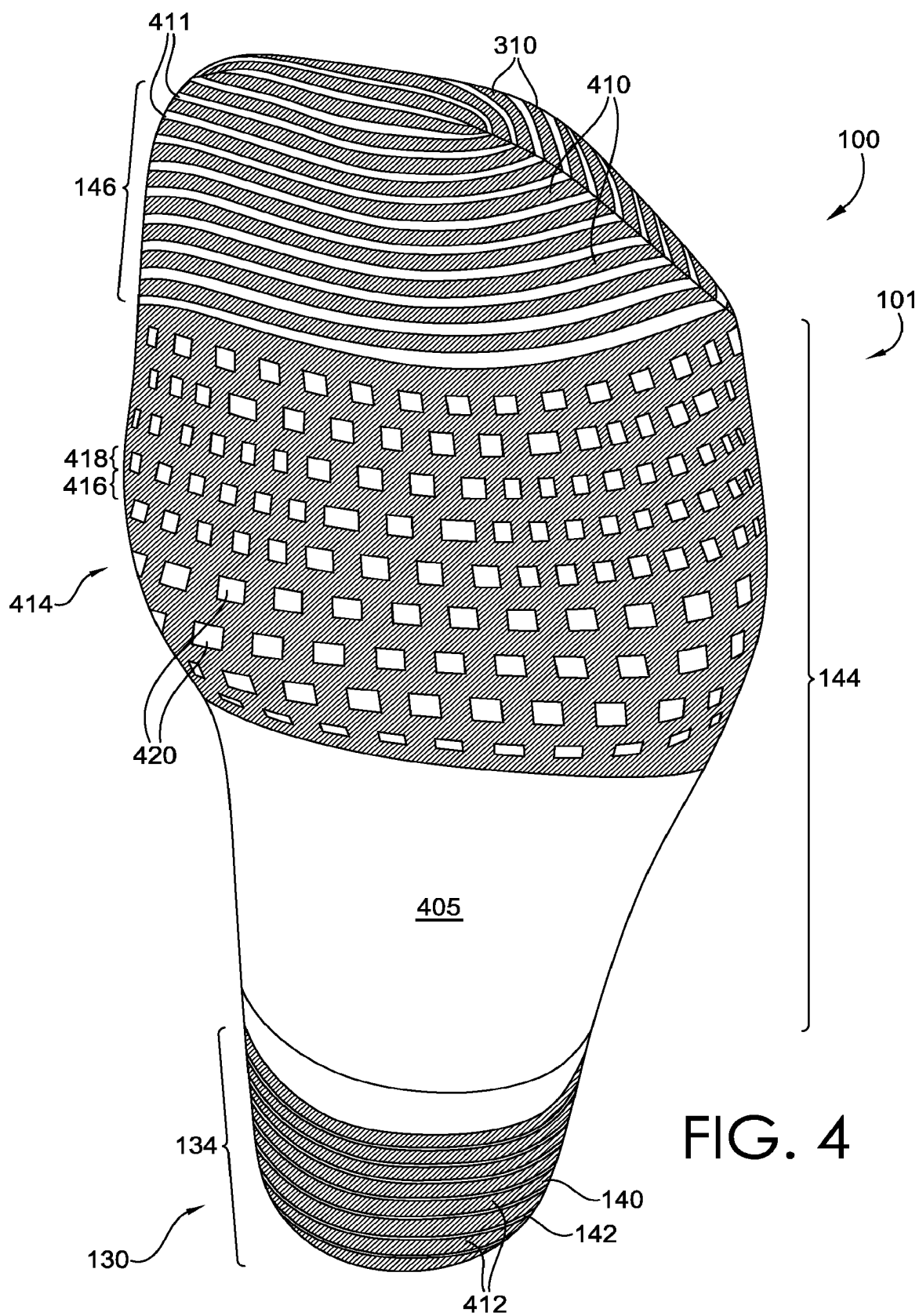
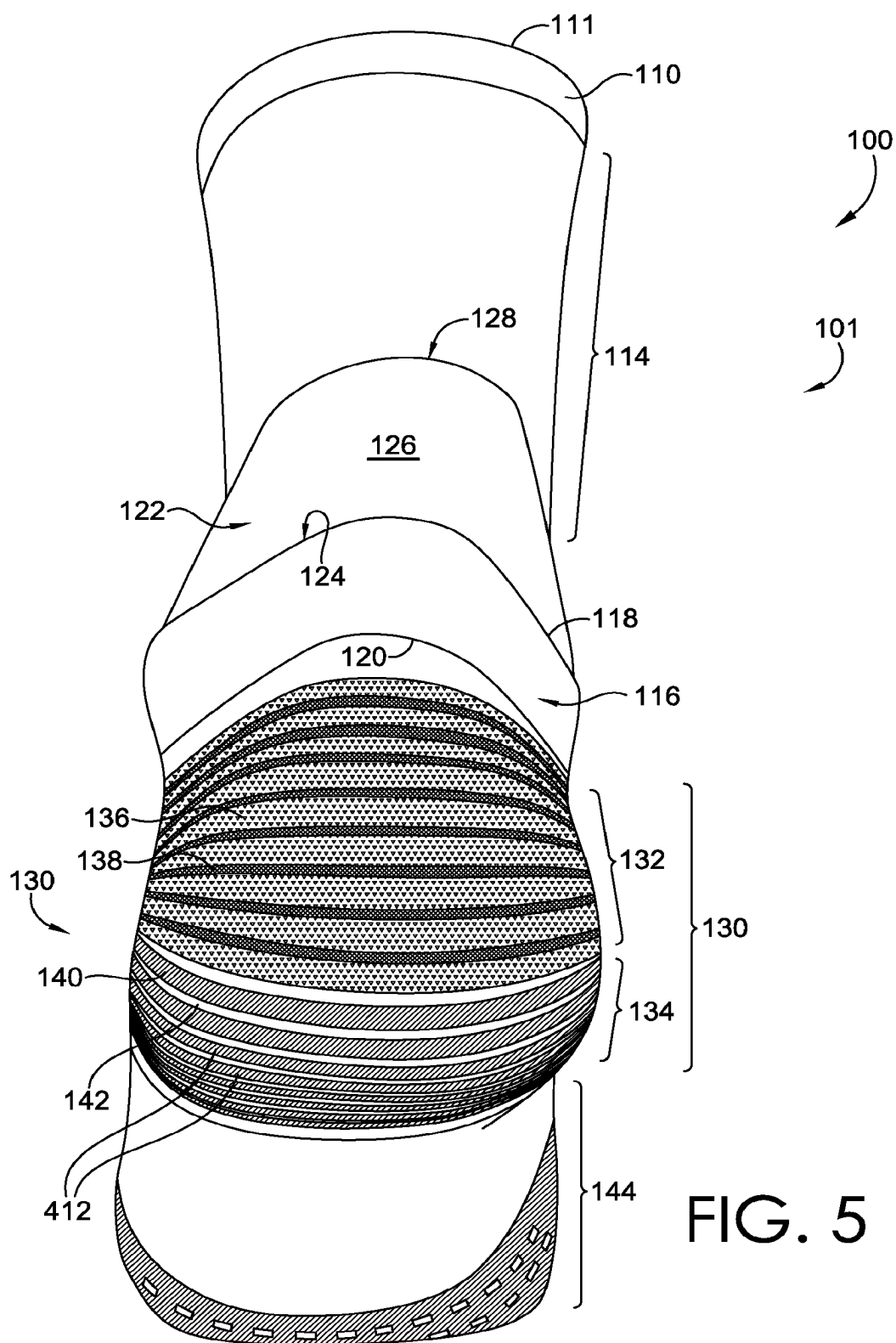


FIG. 1

FIG. 2







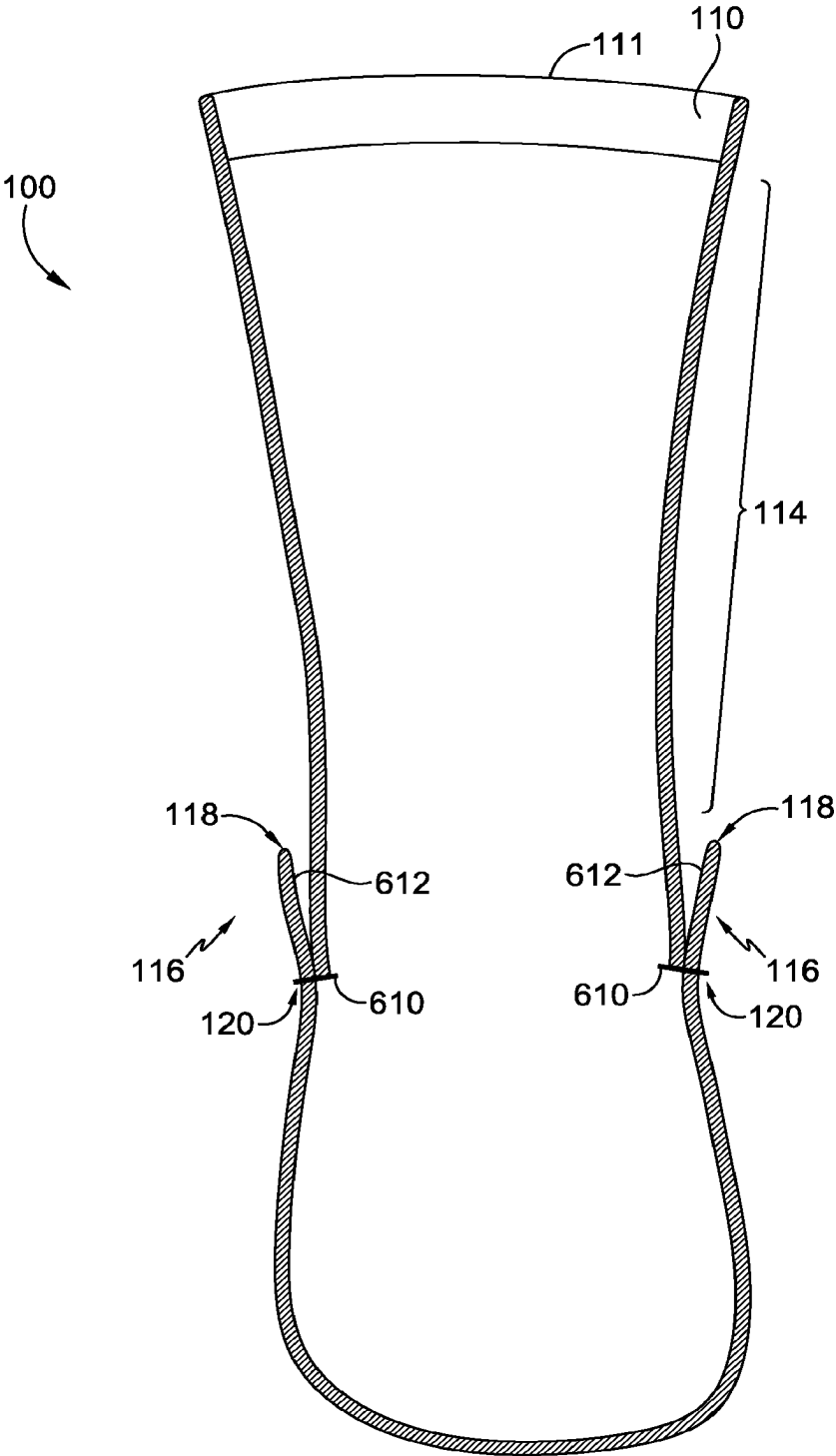
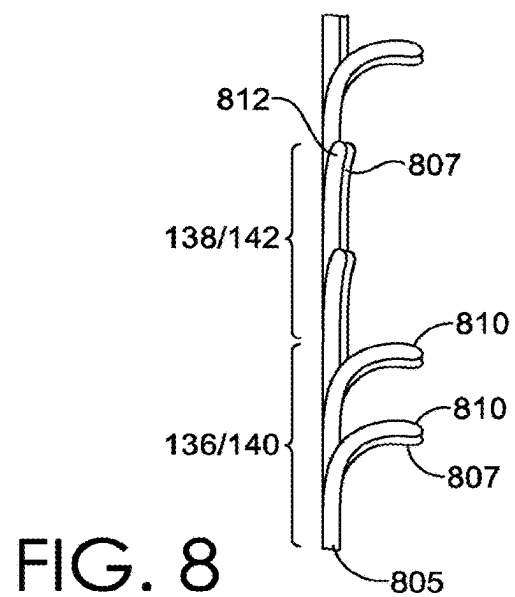
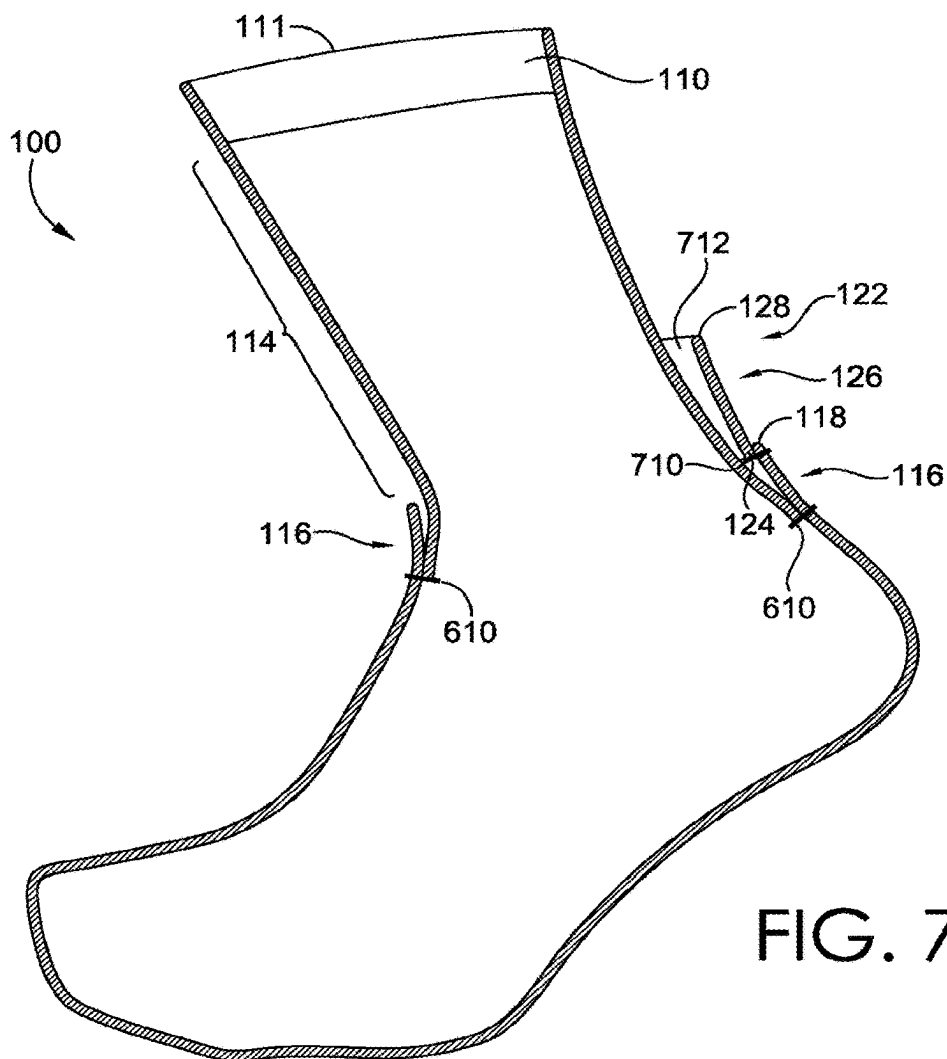
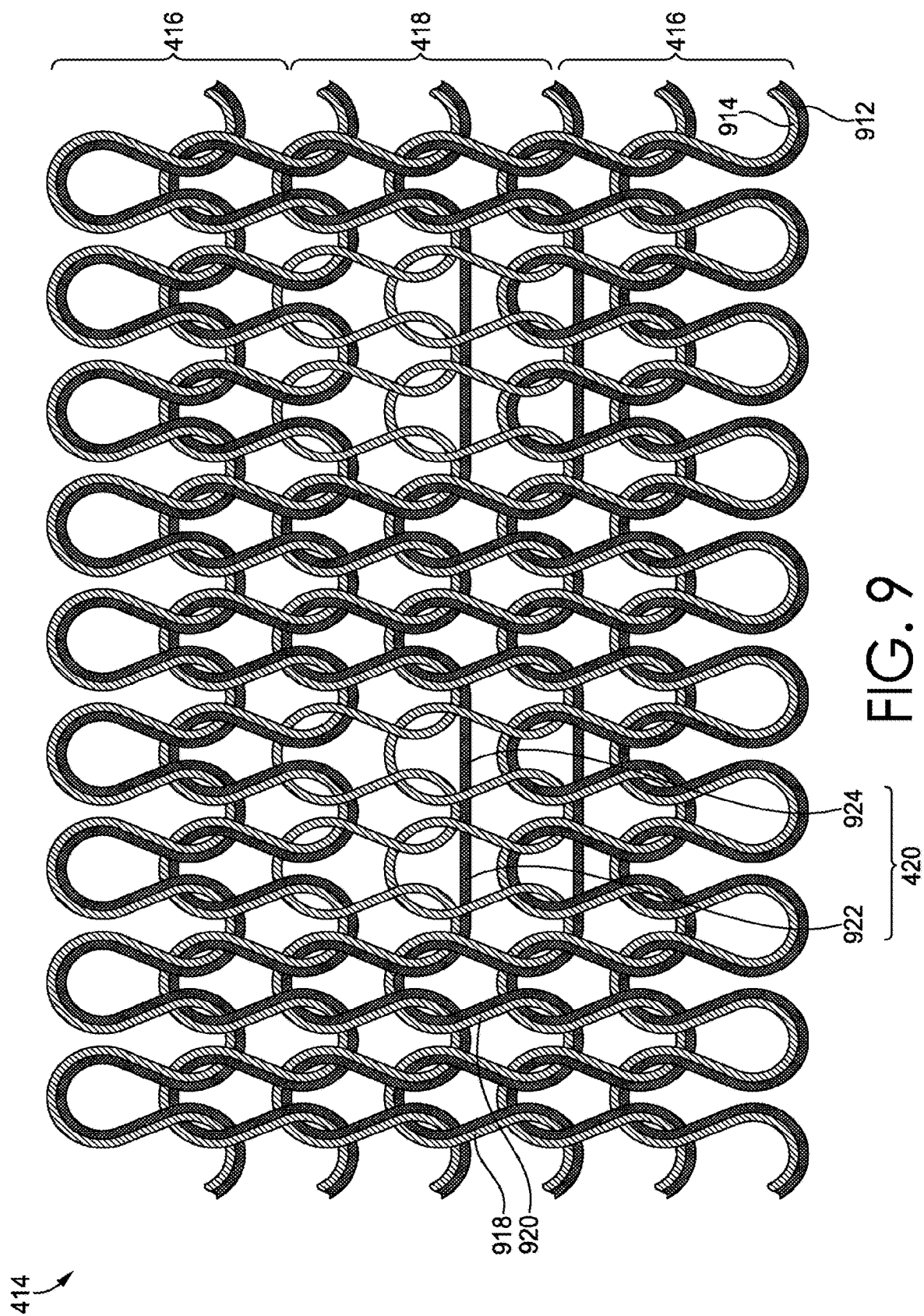


FIG. 6





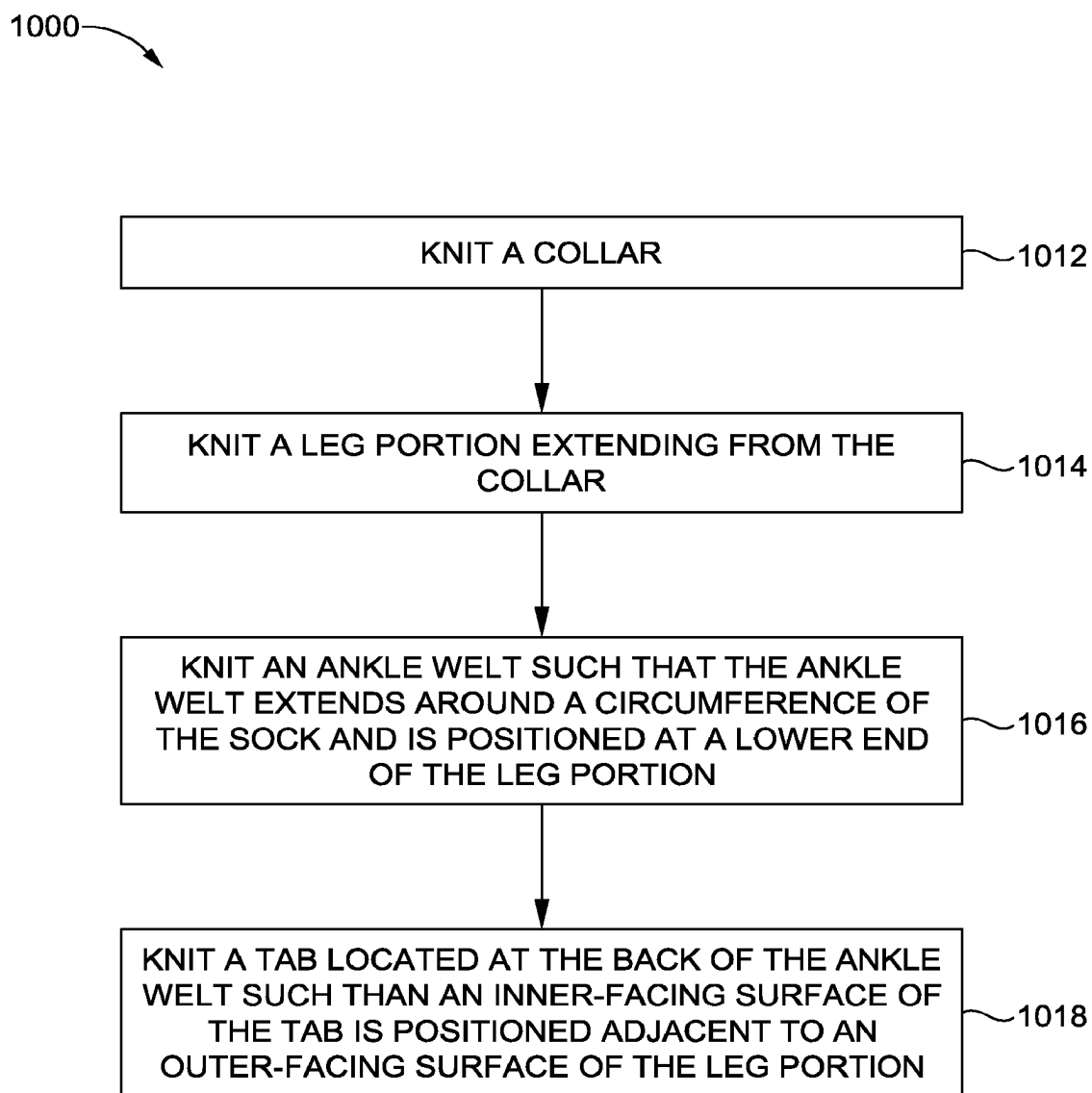


FIG. 10

ATHLETIC SOCK

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application, titled, “Athletic Sock,” is a continuation of co-pending U.S. Non-Provisional patent application Ser. No. 18/530,718, filed Dec. 6, 2023, and titled “Athletic Sock, which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/525,640, filed Nov. 12, 2021 and issued on Jan. 30, 2024 as U.S. Pat. No. 11,882,886, and titled “Athletic Sock,” which claims the benefit of priority of U.S. Application No. 63/157,890, filed Mar. 8, 2021, and titled “Athletic Sock,” U.S. Application No. 63/164,677, filed Mar. 23, 2021, and titled, “Knit Article with Variable Features,” and U.S. Application No. 63/132,593, filed Dec. 31, 2020, and titled “Sleeve for an Extremity.” The entireties of the aforementioned applications are incorporated by reference herein.

TECHNICAL FIELD

[0002] Aspects herein relate to an athletic sock.

BACKGROUND

[0003] Traditional athletic socks may be used for a variety of athletic activities. However, traditional athletic socks may lack features such as zoned cushioning, zoned grip, zoned breathability and permeability, and fit features that make them suitable for certain activities such as running including trail running/hiking. Running, including trail running/hiking, poses a number of challenges such as uneven terrain that causes the foot to shift, for example, medially and laterally within the shoe as well as forward and backward within the shoe, debris (e.g., gravel) that may enter into the space between the sock and the foot especially with ankle length socks, sweating and increased temperatures caused by exertion, and compact surfaces which may impact the joints and muscles of the wearer’s foot.

SUMMARY

[0004] The following clauses represent example aspects of concepts contemplated herein. Any one of the following clauses may be combined in a multiple dependent manner to depend from one or more other clauses. Further, any combination of dependent clauses (clauses that explicitly depend from a previous clause) may be combined while staying within the scope of aspects contemplated herein. The following clauses are examples and are not limiting.

[0005] Clause 1. A knit sock comprising: a collar having a perimeter edge around an opening for receiving a wearer’s foot; a leg portion extending from the collar; a foot portion extending from the leg portion and terminating at a toe end; an integrally knit ankle welt extending around a circumference of the knit sock and positioned at a lower end of the leg portion; and a tab located at a back aspect of the ankle welt, the tab integrally knit with the ankle welt.

[0006] Clause 2. The knit sock according to clause 1, wherein an inner-facing surface of the tab is positioned adjacent to an outer-facing surface of the leg portion.

[0007] Clause 3. The knit sock according to any of clauses 1 through 2, wherein the ankle welt has an upper edge and a lower edge, the lower edge integrally knit with the leg portion.

[0008] Clause 4. The knit sock according to clause 3, wherein the upper edge of the ankle welt is separate from the leg portion.

[0009] Clause 5. The knit sock according to any of clauses 3 through 4, wherein the tab includes a tab base, a tab body extending from the tab base, and a top edge spaced apart from the tab base by the tab body, and wherein the tab base is integrally knit with the upper edge of the ankle welt.

[0010] Clause 6. The knit sock according to clause 5, wherein the top edge of the tab is unattached from the leg portion.

[0011] Clause 7. The knit sock according to any of clauses 1 through 6, further comprising a heel portion positioned between the leg portion and the foot portion, the heel portion having an Achilles section and a plantar section, wherein the Achilles section is formed from a knit structure having an alternating pattern comprising a first number of knit courses knit with terry loops and a second number of knit courses knit without terry loops.

[0012] Clause 8. The knit sock according to clause 7, wherein the first number of knit courses and the second number of knit courses extend from a lateral side to a medial side of the Achilles section.

[0013] Clause 9. The knit sock according to any of clauses 1 through 8, further comprising: a first plurality of integrally knit grip strips located on a lateral side of the foot portion; and a second plurality of integrally knit grip strips located on a medial side of the foot portion.

[0014] Clause 10. The knit sock according to any of clauses 1 through 9, further comprising: a third plurality of integrally knit grip strips located on a dorsal side of the toe end of the knit sock; and a fourth plurality of integrally knit grip strips located on a plantar side of the toe end of the knit sock.

[0015] Clause 11. The knit sock according to any of clauses 1 through 10, further comprising a heel portion positioned between the leg portion and the foot portion, the heel portion having an Achilles section and a plantar section, wherein the plantar section includes a fifth plurality of integrally knit grip strips that extend from a lateral side to a medial side of the plantar part.

[0016] Clause 12. A knit sock comprising: a tubular knit body forming a leg portion and a foot portion extending from the leg portion and terminating at a toe end; an ankle welt extending around a circumference of the tubular knit body and positioned between the leg portion and the foot portion, the ankle welt integrally knit with the tubular knit body; and a tab located at a back aspect of the ankle welt, an inner-facing surface of the tab positioned adjacent to an outer-facing surface of the leg portion.

[0017] Clause 13. The knit sock according to clause 12, wherein the ankle welt has an upper edge and a lower edge, and wherein the tab is integrally knit with the upper edge of the ankle welt.

[0018] Clause 14. The knit sock according to clause 13, wherein the lower edge of the ankle welt is integrally knit with the tubular knit body, and wherein the upper edge of the ankle welt is separate from the leg portion.

[0019] Clause 15. The knit sock according to any of clauses 12 through 14, wherein the tab includes a tab base, a tab body extending from the tab base, and a top edge spaced apart from the tab base by the tab body, wherein: the

tab base is integrally knit with the ankle welt, and each of the tab body and the top edge of the tab are unattached from the leg portion.

[0020] Clause 16. The knit sock according to any of clauses 12 through 15, further comprising a heel portion positioned between the leg portion and the foot portion, the heel portion having an Achilles section and a plantar section, wherein the Achilles section is formed from a knit structure having an alternating pattern comprising a first number of knit courses knit with terry loops and a second number of knit courses knit without terry loops.

[0021] Clause 17. The knit sock according to any of clauses 12 through 16, further comprising one or more integrally knit grip strips that are knit with a yarn formed of filaments having a diameter from about 25 nanometers to about 350 nanometers.

[0022] Clause 18. A method of knitting a sock comprising: during a single knitting event: knitting a collar having a perimeter edge around an opening for receiving a wearer's foot; knitting a leg portion extending from the collar; knitting a foot portion extending from the leg portion and terminating at a toe end; knitting an ankle welt such that the ankle welt extends around a circumference of the sock and is positioned at a lower end of the leg portion; and knitting a tab located at a back aspect of the ankle welt such that an inner-facing surface of the tab is positioned adjacent to an outer-facing surface of the leg portion.

[0023] Clause 19. The method of knitting the sock according to clause 18, further comprising knitting one or more grip strips using a yarn formed of filaments having a diameter from about 25 nanometers to about 350 nanometers.

[0024] Clause 20. The method of knitting the sock according to any of clauses 18 through 19, further comprising knitting a heel portion positioned between the leg portion and the foot portion, the heel portion knit to have an Achilles section and a plantar section, wherein the Achilles section comprises a knit structure having an alternating pattern comprising a first number of knit courses knit with terry loops and a second number of knit courses knit without terry loops.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Examples of aspects herein are described in detail below with reference to the attached drawing figures, wherein:

[0026] FIG. 1 illustrates a lateral view of an example knit sock in accordance with aspects herein;

[0027] FIG. 2 illustrates a medial view of the example knit sock of FIG. 1 in accordance with aspects herein;

[0028] FIG. 3 illustrates a front view of the example knit sock of FIG. 1 in accordance with aspects herein;

[0029] FIG. 4 illustrates a bottom view of the example knit sock of FIG. 1 in accordance with aspects herein;

[0030] FIG. 5 illustrates a rear view of the example knit sock of FIG. 1 in accordance with aspects herein;

[0031] FIG. 6 illustrates a cross-section of the example knit sock taken at cut line 6-6 of FIG. 1 in accordance with aspects herein;

[0032] FIG. 7 illustrates a cross-section of the example knit sock taken at cut line 7-7 of FIG. 3 in accordance with aspects herein;

[0033] FIG. 8 illustrates a knit schematic of a knit structure located at a heel portion of the example knit sock of FIG. 1 in accordance with aspects herein;

[0034] FIG. 9 illustrates a knit schematic of a knit structure located at a forefoot portion of the example knit sock of FIG. 1 in accordance with aspects herein; and

[0035] FIG. 10 illustrates a flow diagram of an example method of manufacturing the example knit sock of FIG. 1 in accordance with aspects herein.

DETAILED DESCRIPTION

[0036] The subject matter of the present invention is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed or disclosed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” might be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly stated.

[0037] Traditional athletic socks may be used for a variety of athletic activities. However, traditional athletic socks may lack features such as zoned cushioning, zoned grip, zoned breathability and permeability, and fit features that make them suitable for certain activities such as running including trail running or hiking. Running, including trail running or hiking, poses a number of challenges such as uneven terrain that causes the foot to shift, for example, medially and laterally within the shoe as well as forward and backward within the shoe, debris (e.g., gravel) that may enter into the space between the sock and the foot especially with ankle length socks, sweating and increased temperatures caused by exertion, and compact surfaces which may impact the joints and muscles of the wearer's foot.

[0038] Aspects herein are directed to a knit sock that includes different knit structures and different yarn types that facilitates its use across a variety of athletic activities including, for example, running. In an example aspect, the knit sock includes a leg portion. The knit sock includes an integrally knit ankle welt that extends around a circumference of the sock at a lower end of the leg portion of the sock. The ankle welt provides 360-degree cushioning at an area corresponding to where a typical shoe collar abuts a wearer's ankle. This may be useful when a wearer is running on uneven terrain causing the foot to shift within the shoe. In an example aspect, a tab is located at a back aspect of the ankle welt, where the tab is integrally knit with the ankle welt. The tab may provide enhanced cushioning along the wearer's Achilles tendon. In addition, the tab may help with donning the sock and may provide an easy way for the wearer to adjust the positioning of the sock if it shifts during wear.

[0039] The knit sock may include additional features that provide zonal cushioning and zonal breathability and/or permeability. For example, the heel portion of the sock may include an Achilles section that is adapted to be positioned adjacent to a wearer's Achilles tendon during wear and a plantar section that is adapted to be positioned adjacent to a wearer's heel during wear. One or more of the Achilles section and the plantar section includes a knit structure

having an alternating pattern of knit courses knit with terry loops to provide cushioning and knit courses knit without terry loops to enhance breathability and permeability of the knit sock.

[0040] The knit sock may also optionally have zoned grip features. For example, integrally knit grip strips knit using a grip yarn may be strategically located to enhance grip where needed. Some example locations include a dorsal side and a plantar side of a toe end of the sock. Engineering the sock to include grip strips in this area may help prevent the toe area of the foot from shifting during wear. This may be especially useful when a wearer is running or hiking down an incline. With traditional socks lacking grip strips in this area, this type of movement may cause the wearer's toes to slide forward and impact the toe end of the shoe, which may cause discomfort and/or bruising of the wearer's toenails. Integrally knit grip strips may also be positioned on the lateral and medial sides of the foot portion of the sock. Positioning grip strips on the medial and lateral sides of the foot portion of the sock may help to minimize the medial to lateral shifting that may occur when a wearer runs/walks on uneven terrain. Grip strips may be absent from the dorsal midfoot and plantar midfoot portions of the sock in example aspects. Not utilizing grip strips on the dorsal midfoot portion of the sock may allow the wearer's skin to shift slightly during wear which helps to reduce tension or shearing forces to this sensitive area. Not utilizing grip strips on the plantar midfoot portion of the sock allows some movement of the sock against the wearer's foot arch, which reduces tension or shearing forces to this sensitive area as well. Integrally knit grip strips may be positioned on the plantar section of the heel portion of the sock to help prevent the heel from shifting especially when running or hiking in hilly terrain.

[0041] In example aspects, the knit sock may optionally include a knit structure at the forefoot of the sock where the knit structure is adapted to be positioned adjacent to the ball of a wearer's foot during wear. The knit structure may provide cushioning, grip, as well as breathability and/or permeability features. In example aspects, the knit structure may include a repeating pattern that has a first number of knit courses of a body yarn and a plating yarn knit in a basic knit stitch with terry loops followed by a second number of knit courses where the body yarn is floated for a first number of stitches and knit for a second number of stitches; the second number of knit courses are integrally knit with the first number of knit courses. In further example aspects, a grip yarn may be used in the knit structure including in the terry loops. Floating the body yarn increases resistance to stretch and increases the allover compression at the forefoot of the sock to provide a close fit. Further, floating the body yarn may increase the breathability and permeability of the knit structure in the float areas since there are fewer knit loops in the float areas which reduces the overall yarn density in these areas. The use of terry loops provides cushioning to this high impact area of the wearer's foot. In addition, using grip yarns in this area helps to stabilize the forefoot within the shoe and thus reduces shifting of the foot during movement.

[0042] The features described above may be used in various combinations, including using all of the features, to provide a knit sock that has zoned grip, zoned cushioning, zoned breathability/permeability, and/or enhanced fit features making the knit sock suitable for athletic activities.

[0043] As used throughout the disclosure, the term "sock" means a knit tubular structure having an open first end and a closed second end. The sock may be worn on a foot of a wearer. The term "inner-facing surface" means the surface of the sock that is configured to face toward a skin surface of a wearer, and the term "outer-facing surface" means the surface of the sock that faces away from the inner-facing surface and toward an external environment. In example aspects, the inner-facing surface may comprise the technical back of the knit tubular structure and the outer-facing surface may comprise the technical front of the knit tubular structure. In example aspects, the knit sock described herein may comprise a crew sock. The term "crew sock" means a sock that includes a leg portion that extends fully or partially over the calf area of a wearer. The sock may be described as having several portions such as a collar, the leg portion, a heel portion, a foot portion, and a toe end. The collar has a perimeter edge that defines an opening for receiving the wearer's foot when the sock is worn. When worn, the collar may be positioned below the wearer's knee in one example. The leg portion extends from the collar and is configured to fully or partially cover the calf/shin area of the wearer.

[0044] The foot portion extends from the leg portion and is configured to cover the wearer's foot. The foot portion, in turn, may include a dorsal side that is configured to cover the dorsum of a wearer's foot, a medial side configured to cover the medial side of the wearer's foot, a lateral side configured to cover the lateral side of the wearer's foot, and a plantar side configured to cover the sole of the wearer's foot. In example aspects, the foot portion may be divided into a toe end configured to cover the toes of the wearer's foot, a forefoot portion that is configured to cover, for example, the area just posterior to the wearer's toes such as the ball of the foot, a midfoot portion configured to cover the middle of the wearer's foot including, for example, the arch area of the wearer's foot, and a hindfoot portion that is configured to cover the posterior part of the wearer's foot. The heel portion of the sock may include a plantar section that covers the plantar surface of the wearer's heel and an Achilles section configured to cover a lower part of the wearer's Achilles tendon. The different areas of the sock described above are intended to represent general areas or proximate locations, are used for reference purposes for the following discussion, and may not be limiting.

[0045] Relational terms used when describing the sock such as upper, lower, inferior, superior, anterior, posterior, front, back, and the like refer to the sock being worn by a wearer as intended with the wearer in a standing position and facing forward.

[0046] The term "knit course" as used herein refers to a predominantly horizontal row of knit loops (in an upright textile as knit) that are produced by adjacent needles during the same knitting cycle. The knit course may comprise one or more stitch types such as a knit stitch, a held stitch, a float stitch, a tuck stitch, a transfer stitch, and the like as these terms are known in the art of knitting. The term "knit stitch" or "basic knit stitch" as used herein refers to the basic stitch type where the yarn is cleared from the needle after pulling a loop of the yarn from the back to the front of the textile through a previous stitch. Thus, the legs of the stitch appear on the technical face of the knit textile and the top and bottom of the stitch appear on the technical back of the textile. A knit stitch or basic knit stitch as used herein may include terry loops. The term "wale" as used herein is a

predominantly vertical column of intermeshed or inter-looped knit loops, generally produced by the same needle at successive (but not necessarily all) courses or knitting cycles. The terms “horizontal” and “vertical” are relative to an upright textile as knit in which the heads of knit loops face toward the top of the textile and the course knit first is oriented toward the bottom of the textile.

[0047] The term “float stitch” as used herein occurs when no new stitch is formed at a needle. Thus, the float yarn may extend across one or more adjacent wales. The term “terry loops” as used herein refers to loops formed from knit yarns that extend away from the technical face and/or the technical back of a knit textile and is to be given the meaning that is commonly used in the art of knitting. In example aspects, the terry loops extend away from the technical back of the knit sock such that they face toward a skin surface of a wearer. The term “plating” as used herein means a knit construction where one or more body yarns and one or more plating yarns are knit in the same knit stitch using, for instance, a body yarn feeder and one or more plating yarn feeders. The term “laid-in yarn” means a yarn that is incorporated into the knit construction during a knitting cycle without the laid-in yarn being knitted into loops.

[0048] Unless otherwise indicated, the body yarn used to knit the knit sock may include a yarn that includes natural fibers/filaments, a yarn that includes man-made fibers/filaments, or a yarn that contains both natural fibers/filaments and man-made fiber/filaments. In one example aspect, the body yarn may include wool fiber/filaments twisted with polyester fiber/filaments. The plating yarn, in example aspects, may include an elastic yarn such as, for example, a spandex yarn covered with nylon. Similarly, the laid-in yarn, in example aspects, may include an elastic yarn such as a LYCRA® yarn covered with nylon.

[0049] The term “welt” as used herein generally refers to a knit structure that may include ribbing and/or may be double knit that is used to support or provide structure to a particular area of the sock. In typical constructions, the welt is generally positioned at the collar of the sock. In aspects contemplated herein, the sock may include a collar welt in addition to a welt positioned at an ankle area of the sock (known as an “ankle welt”) as described further herein.

[0050] The term “integrally knit” as used herein may mean a textile having a yarn from one or more knitted courses being interlooped with one or more knitted courses of another area. For instance, the ankle welt may be integrally knit with, for example, the tubular body of the knit sock if a yarn from the ankle welt is interlooped with a yarn from the tubular body. In another example, the tab may be integrally knit with the ankle welt if a yarn from the tab is interlooped with a yarn from the ankle welt.

[0051] Continuing, it is contemplated herein that the grip strips may be knit with a yarn type having grip characteristics, which is known herein as a “grip yarn.” The term “grip yarns” used when describing yarns that form the grip strips refer to a yarn having a high number of filaments (e.g., 7000 or greater) per single yarn strand such that the denier per filament of the grip yarn is about 0.01 or less. Stated differently, the grip yarn may be formed of filaments having a diameter from about 25 nanometers to about 350 nanometers. In example aspects, the grip yarn may be formed through an “islands-in-the-sea” process. An example filament that may be used in the grip yarn is NANOFRONT® produced by Teijin Limited with headquarters in Tokyo,

Japan. The large number of filaments provides a large surface-to-volume ratio for the yarn, which contributes to the gripping function of the yarn. To describe it differently, the large number of filaments within the yarn causes the yarn to have a higher coefficient of friction as compared to, for example, more typical yarns that incorporate a smaller number of filaments within a single yarn strand such as yarns that incorporate from between, for example, 50 filaments to 500 filaments per yarn. The term “elastic yarn” as used herein refers to the yarn’s ability to stretch from about 100% to about 200% of its original length and recover to approximately (e.g., within 5% to 10%) its original length after the stretching force is removed.

[0052] Unless indicated otherwise, all measurements provided herein are taken when the sock is at standard ambient temperature and pressure (298.15 K and 100 kPa) and the sock is in a resting state (e.g., an unstretched state).

[0053] FIGS. 1 through 5 respectively depict a lateral view, a medial view, a front view, a plantar view, and a rear view of an example knit sock 100 in accordance with aspects herein. The sock 100 is depicted as though worn by a wearer standing with the wearer’s heel elevated to better illustrate different features of the sock 100. The sock 100 generally comprises a tubular body 101 having an open end and a closed end. With respect to FIG. 1, which depicts the lateral view of the knit sock 100, the knit sock 100 includes a collar 110 having a perimeter edge 111 that defines an opening 112 for receiving a wearer’s foot. In some example aspects, the collar 110 may include a welt as shown. A leg portion 114 extends downward from the collar 110. In example aspects, the leg portion 114 may include a plurality of knit courses that extend circumferentially around the leg portion 114.

[0054] The knit sock 100 further includes an ankle welt 116 positioned at a lower end of the leg portion 114 and integrally knit with the leg portion 114. The ankle welt 116 extends around a circumference of the knit sock 100 and includes an upper edge 118 and a lower edge 120. In example aspects, the lower edge 120 is integrally knit with the leg portion 114, and the upper edge 118 is separate from or unattached from the leg portion 114 such that it comprises a free edge. The ankle welt 116 in example aspects may include a ribbed structure and/or may be double knit for extra cushioning, stability, and structure. Positioning the ankle welt 116 such that it extends around an ankle of a wearer when the knit sock 100 is worn and is positioned between a collar of a shoe and the wearer’s ankle provides extra cushioning in this area which may be useful when the foot shifts medially and/or laterally during wear as well as forward or backwards.

[0055] A tab 122 is positioned at a back aspect of the ankle welt 116 and is integrally knit with the ankle welt 116. In example aspects, the tab 122 may be double knit for extra stability and cushioning. In example aspects, the tab 122 may be knit using a high decitex elastic yarn such as a 156 decitex spandex yarn covered with nylon. In example aspects, the tab 122 may not include a plating yarn or a laid-in yarn. Using a high decitex elastic yarn provides structure to the tab 122 as well as stretch and recovery allowing for easier manipulation of the tab 122 by a wearer.

[0056] In an example aspect, the tab 122 includes a tab base 124, a tab body 126 extending from the tab base 124, and a top edge 128 spaced apart from the tab base 124 by the tab body 126 where the top edge 128 is positioned above the tab base 124. In an example aspect, the top edge 128 of the

tab **122** may be positioned from about 3 cm to about 5 cm above the tab base **124**, from about 3.5 cm to about 4.5 cm above the tab base **124**, or about 4 cm above the tab base **124**. As used herein, the term “about” means within $\pm 10\%$ of an indicated value. Configuring the tab **122** to have this height ensures that the top edge **128** is positioned so that it extends above the collar of a typical shoe enabling it to be easily grasped by a wearer when adjusting the sock **100**. In example aspects, the tab base **124** may be integrally knit with the upper edge **118** of the ankle welt **116**. The top edge **128** of the tab **122** may be separate from or unattached from the leg portion **114** of the knit sock **100**. As shown the tab **122** tapers as it extends from the tab base **124** to the top edge **128**.

[0057] Below the ankle welt **116**, the knit sock **100** includes a heel portion **130**. The heel portion **130** includes an Achilles section **132** and a plantar section **134** where the Achilles section **132** is configured to cover the lower end of a wearer's Achilles tendon and the plantar section **134** is configured to cover the plantar portion of the wearer's heel area. In example aspects, and as described further with respect to FIG. 8, the Achilles section **132** has a knit structure that includes an alternating pattern of a first number of knit courses **136** that are knit with terry loops extending from an inner-facing surface of the knit sock **100** and a second number of knit courses **138** that are knit without terry loops. The first number of knit courses **136** and the second number of knit courses **138** extend from a lateral side to a medial side of the Achilles section **132**.

[0058] The first number of knit courses **136** may include a greater number of knit courses than the second number of knit courses **138** such that the ratio of the first number of knit courses **136** to the second number of knit courses **138** may be about 2:1. In one example aspect, the first number of knit courses **136** may include from three knit courses to five knit courses or four knit courses, and the second number of knit courses **138** may include from one knit course to three knit courses or two knit courses.

[0059] In example aspects, the first number of knit courses **136** may be knit with one type of yarn and the second number of knit courses **138** may be knit with a second type of yarn. For instance, the first number of knit courses **136** may be knit with yarns including natural fibers/filaments such as wool, yarns including synthetic fiber/filaments such as polyester, and/or yarns with a mixture of natural and synthetic fibers/filaments. The second number of knit courses **138** may be knit with an elastic yarn such as a spandex yarn covered with polyester or nylon.

[0060] Alternating courses having terry loops with courses not having terry loops provides a desired degree of cushioning while still providing breathability and permeability features. For example, the terry loops provide cushioning, and the areas without terry loops since they have less yarn density, provide breathability and permeability. The ratio of the first number of knit courses **136** to the second number of knit courses **138** may be adjusted as needed to provide more or less cushioning and/or more or less breathability/permeability features. The use of, for instance, wool yarns in the first number of knit courses **136** may contribute to the cushioning features due to the inherent crimp of wool fibers and may also aid in moisture management due to the large moisture regain of wool (e.g., from about 13-18% under standard conditions) which allows the wool yarns to absorb

moisture. The use of elastic yarns in the second number of knit courses **138** provides a snug fit around the wearer's Achilles tendon area.

[0061] The plantar section **134** of the heel portion **130** may include a similar knit structure comprising an alternating pattern of a first number of knit courses **140** knit with terry loops and a second number of knit courses **142** knit without terry loops as described for the Achilles section **132**. As such, the description of the pattern provided with respect to the Achilles section **132** also applies to the plantar section **134**. The first number of knit courses **140** and the second number of knit courses **142** extend from a lateral side to a medial side of the plantar section **134**. In example aspects, similar to the Achilles section, the second number of knit courses **142** may be knit with an elastic yarn such as a spandex yarn covered with polyester or nylon to provide a snug fit. The first number of knit courses **140** may be knit with a grip yarn as explained further below.

[0062] A foot portion **144** extends from the heel portion **130** and the ankle welt **116** to a closed toe end **146** of the knit sock **100**. The leg portion **114** in combination with the foot portion **144** forms the tubular body **101** of the knit sock **100**. The foot portion **144** includes a plurality of knit courses that extend around the circumference of the knit sock **100**. As shown in the lateral view of FIG. 1, the foot portion **144** may include a first plurality of grip strips **148** that are integrally knit with the knit sock **100** using a grip yarn. In example aspects, the first plurality of grip strips **148** are generally positioned on the lateral side of the foot portion **144** of the knit sock **100** such that they do not generally extend on to the dorsal side of the sock or the plantar side of the foot portion **144** of the sock **100**. The first plurality of grip strips **148** may include terry loops extending from an inner-facing surface of the knit sock **100**. Having the grip yarn form terry loops increases the exposed surface area of the grip yarn to provide a greater level of grip. The first plurality of grip strips **148** may help to stabilize and prevent shifting of the wearer's foot within the shoe especially on uneven terrain. The toe end **146** of the knit sock **100** may also include grip strips as further described with respect to FIGS. 3 and 4.

[0063] FIG. 2, which depicts a medial view of the knit sock **100** includes many of the same features (indicated with the same reference numerals) as the lateral view as shown in FIG. 1 but from an opposite side of the knit sock **100**. As such, the ankle welt **116** is shown extending circumferentially around the knit sock **100**, and the tab **122** is shown extending partially around to the medial side of the knit sock **100**. Similarly, the first number of knit courses **136** and the second number of knit courses **138** of the Achilles section **132** are shown extending around the medial side of the heel portion **130**, and the first number of knit courses **140** and the second number of knit courses **142** of the plantar section **134** are shown extending around the medial side of the heel portion **130**.

[0064] As shown in the medial view of FIG. 2, the foot portion **144** may include a second plurality of grip strips **210** that are integrally knit with the knit sock **100** using a grip yarn. In example aspects, the second plurality of grip strips **210** are generally positioned on the medial side of the foot portion **144** of the knit sock **100** such that they do not generally extend on to the dorsal side of the sock or the plantar side of the foot portion **144** of the sock **100**. The second plurality of grip strips **210** may include terry loops extending from an inner-facing surface of the knit sock **100**.

Similar to the first plurality of grip strips **148**, the second plurality of grip strips **210** may help to stabilize and prevent medial and lateral shifting of the wearer's foot within the shoe especially on uneven terrain. By not having the first plurality of grip strips **148** and the second plurality of grip strips **210** extend on to the dorsal side of the sock **100** and the plantar side of at least the midfoot region of the sock **100**, sensitive, thinner skin in this area may move more freely with respect to the inner-facing surface of the knit sock **100** which may lessen shearing or tensioning forces in these areas and improve wearer comfort.

[0065] FIG. 3, which depicts a front view of the knit sock **100**, illustrates many of the same features as shown in FIGS. 1 and 2. As shown in FIG. 3, the first plurality of grip strips **148** and the second plurality of grip strips **210** do not extend on to the dorsal side, indicated by reference numeral **305**, of the foot portion **144** of the knit sock **100**. Stated differently, the first plurality of grip strips **148** and the second plurality of grip strips **210** are generally absent from the dorsal side **305** of the knit sock **100**.

[0066] As shown in FIG. 3, the toe end **146** includes a third plurality of grip strips **310** located on the dorsal side **305** of the sock and extending from the medial side to the lateral side of the toe end **146**. The third plurality of grip strips **310** are integrally knit with the knit sock **100** using a grip yarn. In example aspects, the third plurality of grip strips **310** include terry loops formed from the grip yarn and extending from an inner-facing surface of the knit sock **100**. The third plurality of grip strips **310** may help prevent a wearer's toes from, for example, shifting forward during athletic activities lessening the chances for damage to the wearer's toenails. Adjacent grip strips in the third plurality of grip strips **310** may be separated by knit courses such as those indicated by reference numeral **311** that are knit using a non-grip yarn such as, for example, the body yarn described above.

[0067] FIG. 4 depicts a bottom view of the knit sock **100**. The toe end **146** includes a fourth plurality of grip strips **410** located on a plantar side **405** of the knit sock **100** and extending from the medial side to the lateral side of the toe end **146**. The fourth plurality of grip strips **410** are integrally knit with the knit sock **100** using a grip yarn. In example aspects, the fourth plurality of grip strips **410** include terry loops formed from the grip yarn and extending from the inner-facing surface of the knit sock **100**. The fourth plurality of grip strips **410** in combination with the third plurality of grip strips **310** help to reduce shifting of the wearer's toes during wear. Adjacent grip strips in the fourth plurality of grip strips **410** may be separated by knit courses such as those indicated by reference numeral **411** knit using a non-grip yarn such as, for example, the body yarn described above.

[0068] FIG. 4 further depicts the plantar section **134** of the heel portion **130**. As described in relation to FIG. 1, the plantar section **134** includes a repeating pattern of the first number of knit courses **140** knit with terry loops and the second number of knit courses **142** knit without terry loops. In example aspect, the first number of knit courses **140** are knit with a grip yarn, such that the first number of knit courses **140** form a fifth plurality of grip strips **412**. Positioning the fifth plurality of grip strips **412** on the plantar section **134** of the heel portion **130** helps to lock the wearer's heel in place and prevent shifting during, for example, walking or running on uneven terrain.

[0069] FIG. 4 additionally depicts a knit structure, referenced generally by the numeral **414**, that is located on the plantar side **405** of a forefoot portion of the foot portion **144** directly posterior to the fourth plurality of grip strips **410** at the toe end **146**. When the knit sock **100** is worn, the knit structure **414** is positioned adjacent to the ball of the wearer's foot. As will be described further with respect to FIG. 9, the knit structure **414** may include a repeating pattern that has a first number of knit courses **416** of a body yarn and a plating yarn knit in a basic knit stitch with terry loops followed by a second number of knit courses **418** where the body yarn is floated for a first number of stitches and knit for a second number of stitches. The areas of the knit structure **414** that include the floats are indicated by reference numeral **420**. Floating the body yarn increases resistance to stretch and increases the allover compression at this portion of the knit sock **100**. Use of the terry loops in the knit structure **414** provides cushioning to this high impact area of the wearer's foot. In addition, because the float areas **420** in the second number of knit courses **418** have a lower yarn density than, for example, areas comprising knit stitches without floats, breathability and permeability in the float areas **420** may be enhanced.

[0070] FIG. 5 depicts a rear view of the knit sock **100**. As shown, the tab **122** is positioned at the back aspect of the knit sock **100** and extends from a lateral side to a medial side of the knit sock **100**. Such positioning allows easy grasping of the tab **122** by a wearer to provide fit adjustment of the knit sock **100** particularly at the ankle area. The Achilles section **132** of the heel portion **130** is depicted with the repeating pattern of the first number of knit courses **136** that include terry loops and the second number of knit courses **138** that do not include terry loops. The plantar section **134** of the heel portion **130** includes the repeating pattern of the first number of knit courses **140** knit with terry loops and the second number of knit courses **142** knit without terry loops where the first number of knit courses **140** are knit with the grip yarn and form the fifth plurality of grip strips **412**.

[0071] FIG. 6 is a cross-section of the knit sock **100** taken at cut line 6-6 of FIG. 1. FIG. 6 is provided to illustrate how the ankle welt **116** is integrally knit with the knit sock **100** and with the leg portion **114** of the knit sock **100**. The lower edge **120** of the ankle welt **116** is integrally knit with the lower end of the leg portion **114** as indicated by the same shading pattern used throughout. Reference numeral **610** schematically depicts a yarn that is interlooped with one or more yarns of the leg portion **114** and one or more yarns of the ankle welt **116**. The upper edge **118** of the ankle welt **116** is unattached from or separate from the leg portion **114**, and an inner-facing surface **612** of the ankle welt **116** is positioned adjacent and/or faces toward the leg portion **114**.

[0072] FIG. 7 is a cross-section of the knit sock **100** taken at cut line 7-7 of FIG. 3. FIG. 7 is provided to illustrate how the tab **122** is integrally knit with the ankle welt **116**. The tab base **124** is integrally knit with the upper edge **118** of the ankle welt **116**. Reference numeral **710** schematically depicts a yarn that is interlooped with one or more yarns of the ankle welt **116** and one or more yarns of the tab **122**. The top edge **128** of the tab **122** is unattached from or separate from the leg portion **114**, and an inner-facing surface **712** of the tab **122** is positioned adjacent and/or faces toward the leg portion **114**.

[0073] FIG. 8 is an enlarged schematic sectional view illustrating a portion of the knit structure forming, for

example, the Achilles section **132** and/or the plantar section **134** of the heel portion **130**. The knit structure includes the first number of knit courses **136/140** that are knit with terry loops as indicated by reference numeral **810** and the second number of knit courses **138/142** knit without the terry loops **810**. Although not shown, the first number of knit courses **136/140** may include a greater number of knit courses than the second number of knit courses **138/142** such that the ratio of the first number of knit courses **136/140** to the second number of knit courses **138/142** may be about 2:1.

[0074] The first number of knit courses **136/140** are knit with a body yarn **805** and a plating yarn **807**. With respect to the first number of knit courses **136**, the body yarn **805** may include a yarn with natural fibers/filaments such as wool, synthetic fiber/filaments such as polyester, and/or a mixture of natural and synthetic fibers/filaments. With respect to the first number of knit courses **140**, the body yarn **805** may include a grip yarn. The plating yarn **807** in the first number of knit courses **136** and the first number of knit courses **140** may include an elastic yarn such as a spandex yarn covered with polyester or nylon.

[0075] In some example aspects, the second number of knit courses **138/142** may be knit with a different body yarn **812** than the body yarn **805**. For example, with respect to both the second number of knit courses **138** and the second number of knit courses **142**, the body yarn **812** may include an elastic yarn such as a spandex yarn covered with polyester or nylon. The same plating yarn **807** may be used for both the second number of knit courses **138** and the second number of knit courses **142**.

[0076] FIG. 9 depicts a knit schematic of the knit structure **414** that is located on the plantar side **405** of a forefoot portion of the foot portion **144** directly posterior to the fourth plurality of grip strips **410** at the toe end **146**. When the knit sock **100** is worn, the knit structure **414** is positioned adjacent to the ball of the wearer's foot. It is contemplated herein that the knit structure **414** may be used in other locations on the knit sock **100**.

[0077] The knit structure **414** includes the first number of knit courses **416** knit with a basic knit stitch using at least a body yarn **912** (shown with cross-hatching) and a plating yarn **914** (shown with hatching). In example aspects, the body yarn **912** may include a grip yarn. Although terry loops are not shown, it is contemplated herein that one or more of the body yarn **912** and the plating yarn **914** in the first number of knit courses **416** may be knit to form terry loops on the technical back of the knit structure **414** to provide cushioning as is known in the art of knitting. The first number of knit courses **416** may include two courses as shown. In example aspects, the first number of knit courses **416** may include more than two knit courses such as three knit courses, four knit courses, and the like. Or the first number of knit courses **416** may include a fewer number of knit courses than shown, such as one knit course. The selection of the number of knit courses in the first number of knit courses **416** may be based on desired end properties of the knit structure **414** such as, for example, increased or decreased cushioning.

[0078] Next, the knit structure **414** includes the second number of knit courses **418** that have a repeating pattern of two basic knit stitches, such as knit stitches **918** and **920** knit with the body yarn **912** and the plating yarn **914**, followed by two float stitches, such as float stitches **922** and **924** to form the float areas **420**. In this example aspect, the body

yarn **912** is floated across two wales while the plating yarn **914** continues to be knit with a basic knit stitch, followed by two basic knit stitches knit with the body yarn **912** and the plating yarn **914**, followed by two float stitches, and so on. Floating at least the body yarn **912** increases the stretch resistance of the knit structure **414** (the knit structure **414** is more resistant to stretch) because there is less yarn available for stretching as compared to a knitted loop. This, in turn, increases the compression of the knit structure **414** which may create a tighter fit at the forefoot portion of the foot portion **144** and prevent shifting or rotating of the knit sock **100**. Further, due to decreased yarn density at the float areas **420**, breathability and/or permeability in these areas may be increased. Knitting the plating yarn **914** in the float areas **420** increases the structural stability of the knit structure **414**.

[0079] Although the second number of knit courses **418** is shown comprising two knit courses, it is contemplated herein that the second number of knit courses **418** may include additional knit courses such as three knit courses, four knit courses, and the like. It is further contemplated herein that the second number of knit courses **418** may include the same number of knit courses as the first number of knit courses **416** as shown in FIG. 9. It is also contemplated herein that the second number of knit courses **418** may comprise a fewer number of knit courses than the first number of knit courses **416** or a greater number of knit courses than the first number of knit courses **416**. The selection of the number of knit courses in the second number of knit courses **418** may be based on desired end properties of the knit structure **414** such as, for example, increased or decreased resistance to stretch and/or breathability.

[0080] It is also contemplated herein that the body yarn **912** may extend over a fewer number of wales or of a greater number of wales than that shown. By extending the body yarn **912** over a fewer number of wales (e.g., one wale), the resistance to stretch and/or breathability may be decreased, and by extending the body yarn over a greater number of wales (e.g., greater than two wales), the resistance to stretch and/or breathability may be increased. It is additionally contemplated that the number of float stitches (two in this example) may differ from the number of basic knit stitches that immediately precede and/or follow the float stitches. Any and all aspects, and any variation thereof are contemplated as being within aspects herein.

[0081] The knit structure **414** is a repeating pattern such that the first number of knit courses **416** are repeated as shown in FIG. 9. The repeating pattern thus comprises a first number of knit courses having a basic knit stitch with terry loops, and a second number of knit courses where the body yarn is knit for a first number of stitches and then floated for a second number of stitches.

[0082] FIG. 10 is a flow diagram of an example method **1000** of manufacturing a knit sock such as the knit sock **100**. The method **1000** is executed during a single knitting event on a knit machine such as, for example, a circular knit machine configured to knit socks. At a step **1012**, a collar, such as the collar **110** is knit where the collar has a perimeter edge around an opening for receiving a wearer's foot. At a step **1014**, a leg portion, such as the leg portion **114** is knit where the leg portion extends from the collar. At a step **1018**, an ankle welt, such as the ankle welt **116** is knit so that the ankle welt extends around a circumference of the knit sock and is positioned at a lower end of the leg portion. At a step **1020**, a tab, such as the tab **122** is knit at a back aspect of

the ankle welt. In example aspects, an inner-facing surface of the tab is positioned adjacent to an outer-facing surface of the leg portion.

[0083] The method **1000** may include additional steps such as knitting a foot portion (e.g., the foot portion **144**) where the foot portion extends from the leg portion. The method **1000** may further include knitting one or more grip strips using a yarn formed of filaments having a diameter from about 25 nanometers to about 350 nanometer. In addition, the method **1000** may include knitting a heel portion, such as the heel portion **130**, where the heel portion is located between the leg portion and the foot portion. The heel portion is knit to have an Achilles section, such as the Achilles section **132**, and a plantar section, such as the plantar section **134**. One or more of the Achilles section and the plantar section may include a knit structure having an alternating pattern comprising a first number of knit courses (e.g., first number of knit courses **136** and **140**) knit with terry loops and a second number of knit courses (e.g., second number of knit courses **138** and **142**) knit without terry loops.

[0084] The knitting steps described above may occur in the order listed but it is contemplated herein that the knitting steps may occur in a different order. For example, a toe end of the knit sock may be knit first.

[0085] Aspects of the present disclosure have been described with the intent to be illustrative rather than restrictive. Alternative aspects will become apparent to those skilled in the art that do not depart from its scope. A skilled artisan may develop alternative means of implementing the aforementioned improvements without departing from the scope of the present disclosure.

[0086] It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations and are contemplated within the scope of the claims. Not all steps listed in the various figures need be carried out in the specific order described.

1. A knit sock comprising:
 - a collar having a perimeter edge around an opening for receiving a wearer's foot;
 - a leg portion extending from the collar;
 - a foot portion extending from the leg portion and terminating at a toe end; and
 - a heel portion positioned between the leg portion and the foot portion, the heel portion having an Achilles section and a plantar section;
 wherein at least the Achilles section comprises a knit structure having an alternating pattern comprising a first number of knit courses knit with terry loops and a second number of knit courses knit without terry loops.
2. The knit sock of claim 1, wherein the first number of knit courses and the second number of knit courses extend from a lateral side to a medial side of the Achilles section.
3. The knit sock of claim 2, wherein the first number of knit courses greater than the second number of knit courses.
4. The knit sock of claim 3, wherein the ratio of the first number of knit courses to the second number of knit courses is about 2:1.
5. The knit sock of claim 1, wherein the plantar section includes a first plurality of integrally knit grip strips that extend from a lateral side to a medial side of the plantar section.

6. The knit sock of claim 5, wherein the first plurality of integrally knit grip strips comprises terry loops.

7. The knit sock of claim 6, further comprising:

- a second plurality of integrally knit grip strips located on a lateral side of the foot portion; and
- a third plurality of integrally knit grip strips located on a medial side of the foot portion.

8. The knit sock of claim 7, further comprising:

- a fourth plurality of integrally knit grip strips located on a dorsal side of the toe end of the knit sock; and
- a fifth plurality of integrally knit grip strips located on a plantar side of the toe end of the knit sock.

9. The knit sock of claim 8, wherein the integrally knit grip strips are knit with a yarn comprising filaments having a diameter from about 25 nanometers to about 350 nanometers.

10. The knit sock of claim 6, wherein the integrally knit grip strips are knit with a grip yarn.

11. The knit sock of claim 10, wherein the second number of knit courses comprises an elastic yarn.

12. A knit sock comprising:

- a collar having a perimeter edge around an opening for receiving a wearer's foot;
- a leg portion extending from the collar;
- a foot portion extending from the leg portion and terminating at a toe end; and
- a heel portion positioned between the leg portion and the foot portion, the heel portion having an Achilles section and a plantar section;

wherein:

- the Achilles section comprises a first knit structure having an alternating pattern comprising a first number of knit courses knit with terry loops and a second number of knit courses knit without terry loops; and
- the plantar section comprises a second knit structure having an alternating pattern comprising a third number of knit courses knit with terry loops and a fourth number of knit courses knit without terry loops.

13. The knit sock of claim 12, wherein the third set of knit courses comprises a first body yarn and a first plating yarn.

14. The knit sock of claim 13, wherein the fourth set of knit courses comprises a second body yarn and a second plating yarn.

15. The knit sock of claim 14, wherein the first body yarn and the second body yarn are a different type of yarn.

16. The knit sock of claim 15, wherein the second body yarn and the second plating yarn comprise an elastic yarn.

17. A knit sock comprising:

- a collar having a perimeter edge around an opening for receiving a wearer's foot;
- a leg portion extending from the collar;
- a foot portion extending from the leg portion and terminating at a toe end; and
- a heel portion positioned between the leg portion and the foot portion, the heel portion having an Achilles section and a plantar section;

wherein:

- the Achilles section comprises a knit structure having an alternating pattern comprising a first number of knit courses knit with terry loops and a second number of knit courses knit without terry loops;
- the plantar section comprises a second knit structure having an alternating pattern comprising a third

number of knit courses knit with terry loops and a fourth number of knit courses knit without terry loops;

the first number of knit courses is knit with a first body yarn and a first plated yarn,

the second number of knit courses is knit with a second body yarn and a second plated yarn;

the third number of knit courses is knit with a third body yarn and a third plated yarn; and

the fourth number of knit courses is knit with a fourth body yarn and a fourth plated yarn.

18. The knit sock of claim **17**, wherein the first plating yarn, the second plating yarn, the third plating yarn and the fourth plating yarn comprise an elastic yarn.

19. The knit sock of claim **18**, wherein the first body yarn is different than the second body yarn.

20. The knit sock of claim **19**, wherein the third body yarn is a grip yarn.

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